Use three separate files:

```
structured_data.csv
semi_structured_data.json
unstructured_data.txt
```

1. Structured Data — CSV

Create:

```
structured_data.csv
```

Paste:

```
Shipment_ID,Origin,Destination,Mode,Weight_Kg,Distance_Km,Delivery_Days,C
S1001,Ahmedabad,Surat,Road,120,265,1,4200
S1002,Pune,Nashik,Road,85,210,1,3100
S1003,Chennai,Kochi,Rail,300,690,2,7800
S1004,Delhi,Jaipur,Road,150,280,1,4600
S1005,Bengaluru,Hyderabad,Rail,420,575,2,9200
S1006,Indore,Bhopal,Road,95,190,1,2900
S1007,Mumbai,Goa,Air,60,590,1,11500
S1008,Patna,Ranchi,Road,110,330,2,3900
S1009,Kolkata,Bhubaneswar,Rail,250,440,2,6800
S1010,Hyderabad,Vijayawada,Road,180,275,1,5100
```

Create:

```
structured_assignment.py
```

Use Pandas.

Assignments

1. Load the CSV into a DataFrame.
2. Display only:

- Shipment_ID
- Origin
- Destination
- Cost

3. Show all shipments where `Mode` is `Road`.

4. Show shipments with `Weight_Kg` greater than 150.

5. Create a new column:

```
Cost_Per_Km = Cost / Distance_Km
```

6. Find:

   - average shipment cost
   - highest shipment cost
   - lowest shipment cost

7. Group by `Mode` and calculate total cost.

8. Group by `Mode` and calculate average weight.

9. Sort shipments from highest to lowest cost.

10. Display shipments where:

- Delivery_Days <= 1
- and Cost < 5000

---

2. Semi-Structured Data — JSON

Create:

```
semi_structured_data.json
```

Paste:

```
[
  {
    "incident_id": "INC201",
```

```
  "location": {
    "city": "Pune",
    "site": "Warehouse A"
  },
  "severity": "High",
  "category": "Equipment",
  "reported_by": "Supervisor",
  "actions": ["Machine stopped", "Area isolated"]
},
{
  "incident_id": "INC202",
  "location": {
    "city": "Nagpur",
    "site": "Distribution Hub"
  },
  "severity": "Medium",
  "category": "Safety",
  "reported_by": "Technician"
},
{
  "incident_id": "INC203",
  "location": {
    "city": "Chennai",
    "site": "Plant 2"
  },
  "severity": "Low",
  "category": "Maintenance",
  "reported_by": "Engineer",
  "actions": ["Inspection scheduled"]
},
{
  "incident_id": "INC204",
  "location": {
    "city": "Vadodara",
    "site": "Plant 1"
  },
  "severity": "High",
  "category": "Safety",
  "reported_by": "Manager",
  "actions": ["Operations paused", "Safety team alerted"]
},
{
  "incident_id": "INC205",
  "location": {
```

```
      "city": "Mysuru",
      "site": "Warehouse C"
    },
    "severity": "Medium",
    "category": "Equipment",
    "reported_by": "Operator"
  }
]
```

Create:

```
semi_structured_assignment.py
```

Assignments

1. Load the JSON file using Python.
2. Print all incident IDs.
3. Print the city of each incident.
4. Display incidents where `severity` is `High`.
5. Display incidents where `category` is `Safety`.
6. Some incidents do not contain `actions`. Print:

```
No actions recorded
```

when the field is missing.

7. Count how many incidents belong to each severity level.
8. Add a new field:

```
status = "Open"
```

to every incident.

9. Save the updated records to:

```
updated_incidents.json
```

10. Convert the JSON into a Pandas DataFrame using:

```
pd.json_normalize()
```

and display:

- incident_id
- severity
- category
- location.city
- location.site

---

3. Unstructured Data — Text File

Create:

```
unstructured_data.txt
```

Paste:

```
During the morning inspection, the operations team noticed an unusual sou

The maintenance team was informed immediately and the conveyor was tempor

A technician found that one of the rollers was damaged and required repla

The nearby loading area remained operational, although movement was slowe

Later in the afternoon, the damaged roller was replaced and the conveyor

The team reported that normal operations resumed at 4:30 PM.

No injuries were reported during the incident.

A follow-up inspection has been scheduled for next week.
```

Create:

```
unstructured_assignment.py
```

Assignments

1. Read and print the complete text file.
2. Count the total number of words.
3. Count the total number of lines.
4. Check whether the word:

```
maintenance
```

appears in the report.

5. Count how many times the word:

```
conveyor
```

appears.

6. Print only lines containing:

```
team
```

7. Print only lines containing:

```
incident
```

8. Find and print the sentence that mentions the damaged component.
9. Convert the complete text to lowercase and save it as:

```
lowercase_report.txt
```

10. Create a simple summary showing whether these keywords appear:

```
maintenance
injuries
inspection
replacement
operations
```

Expected format:

```
maintenance : Found
injuries    : Found
inspection  : Found
replacement : Found
operations  : Found
```
